CivicTrack System Architecture and Experimental Workflow

Condition A: Tabular List Explorer

Conventional Baseline (LPSE / Satu Data Style)

- HTML5 / React 18 Table Component
- Linear Data Grid (Code, Name, Budget, Dates)
- Textual Progress & Calendar Percentages
- Search Input & Kecamatan Dropdown Filter
- Standard Status Text Pills (In Progress, Selesai)

Condition B: Map & Timeline Explorer

CivicTrack Integrated Visual Interface

- Leaflet.js + GeoJSON Administrative Polygons
- Interactive Spatial Clusters & Color-Coded Markers
- Synchronized Dual Gantt Milestone Bars (EVM)
- Contractual Deadline vs. Field Update Timestamps
- Dynamic Staleness Badges & Delay Tags

Analytical Engine & Business Logic

Earned Value Management (EVM) Core

- Time Elapsed Engine: T_elap = min(100, (T_ref-T_start)/(T_end-T_start))
- Milestone Physical Progress: P_physical = sum(w_k * c_k)
- Schedule Variance: Delta = P_physical - T_elapsed
- Strict Priority Classifier (Overdue, Delayed, On-Track)
- Staleness Detection Engine (Alert when Delta_t > 90 days)

FastAPI Backend & Geospatial Services

Server Layer & Telemetry Engine

- Python 3.11 High-Performance REST API
- Strict Pydantic v2 Schema Validation
- GeoJSON Spatial Boundary Endpoints (27 Kecamatans)
- Project Repository & Query Optimization
- Audit Logging & Data Integrity Safeguards

ISO/IEC/IEEE 29119 Software Testing & ISO/IEC 25010 Quality Standards

Automated Verification: 38 Backend Pytest Suites + 57 Frontend Vitest Suites (100% Pass Rate across 95 Test Cases)